

Shayan Shakeri

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PROFESSIONAL SUMMARY

AI research engineer focused on evaluating LLM and agent systems: eval pipelines and infrastructure that detect misaligned outputs at scale. At BMO's AI Centre of Excellence, built a deterministic agent eval pipeline that surfaced a systematic risk-downplaying bias in a GenAI tool serving \$200B+ AUM. Co-founded Merlyn Labs: VLM judges built to resist reward gaming, an 8th-place finish in Stanford's BEHAVIOR-1K Challenge, and published failure-mode analyses (merlyn-labs.com/behavior-report).

CORE COMPETENCIES

LLM & Agent Evals

Eval Infrastructure

Agent Scaffolds

VLM Judges

Technical Writing

WORK EXPERIENCE

BMO AI Centre of Excellence Sep 2025 - Present

AI Research Engineer | Toronto, Canada

- Uncovered systematic bias to downplay investment risk in a GenAI tool serving **\$200B+ AUM** wealth management
- Built a **deterministic agent eval pipeline** using hundreds of synthesized inputs to detect misaligned outputs at scale
- Building a **graph-based agentic system** answering complex multi-hop relational queries across bank client data

Merlyn Labs Aug 2025 - Present

AI Researcher & Co-Founder | Toronto, Canada

- Developing **VLM judges** that score rollouts into dense, context-dependent RL rewards that are difficult to game
- Open-sourced **flow-matching VLA integration for RLinf**, enabling RL training on the BEHAVIOR-1K suite in OmniGibson
- Showed $\pi 0.5$ generalization failures on LIBERO-PRO are recipe-induced overfitting; proposed an alternative baseline
- Conservative finetuning doubled position-swap success (**21% to 42%**), showing spatial generalization over memorization
- Running sim-to-real transfer of household tasks on hand-built AlohaMini embodiment

Epineuron Technologies May 2021 - Aug 2022

Biomedical & Electrical Engineering Co-op | Mississauga, Canada

- Helped develop **PeriPulse**, an FDA Breakthrough-designated device now in multi-national clinical trials
- Designed and assembled **PCBs** and authored validation protocols (IEC 60601-1, ISO 13485) for clinical test runs
- Optimized power draw via oscilloscope/power analyzer testing, achieving **900%** battery life improvement
- Modeled electromagnetic field penetration into peripheral nerves in **COMSOL**; data used to optimize electrode diameter

RESEARCH & PROJECTS

Stanford BEHAVIOR-1K Challenge 8th Place | Standard Track Sep 2025 - Nov 2025

- Identified **proprioceptive collapse** as critical VLA failure mode; 60% masking improved task success by up to **48%**
- Found chunked execution outperformed temporal ensembling by **3x**, revealing VLA architectures lack temporal awareness
- Trained on 10,000+ demonstrations across 22 tasks; published a [technical report](#) and further analysis on [LessWrong](#)

EDUCATION

M.Eng, MIE - AI & Robotics | University of Toronto Sep 2024 - Apr 2027 (anticipated)

Coursework: Deep Learning, Reinforcement Learning, AI Applications in Robotics, Healthcare Robotics

B.Eng, Engineering Physics (Co-op) | McMaster University Sep 2018 - Jun 2023

Dean's Honour List. Coursework: Computational Multiphysics, Quantum Computing, Numerical Methods, Signals & Systems

SKILLS & INTERESTS

ML & AI: Python, C/C++, PyTorch, JAX, RL, Imitation Learning, LLM Fine-tuning, Alignment Evaluation, Agents

Robotics & Hardware: VLA Models, OmniGibson, MuJoCo, Flow Matching, Sim2Real, ROS2, PCB Design, 3D Printing

Interests: Literature, Dungeons & Dragons, Volleyball, Twilight Imperium, Piano